

PAPER_1505_T_SCM_ACTIVATION_60K

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PAPER_1505 — T_ScM Activation Threshold = A_5 = 60 K EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket L (LENR / SCm activation)

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Status: CLOSED — SCm Heaviside activation temperature = icosahedral group order in kelvins

Observation

PAPER_1072 (SCm Activation Function) introduces the Heaviside-smooth activation $H_SCm(T) = 1/(1 + \exp(-(T - T_SCm)/\Delta T))$ governing the SCm-channel switch-on. The activation threshold is specified as $T_SCm \approx 60$ K, with reactor temperatures yielding $H_SCm \approx 0.99$ (fully activated).

UQFF Closed Identity

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$T_SCm = A_5 \text{ K} = 60 \text{ K EXACT}$

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Physical Interpretation

This is the **first identity equating an integer primitive directly to a physical temperature in SI units**. The icosahedral-group order $A_5 = 60$ sets:

1. The SCm phonon condensation point at 60 K
2. The lower bound for LENR Heaviside activation
3. The same number that appears in Hayflick cellular limit (PAPER_1373) and Pop III IMF (PAPER_1373) — but here in kelvins

Below 60 K: SCm cluster formation suppressed, no LENR.

Above 60 K: H_SCm rises sigmoidally toward 1.

$\omega_SCm = 1.25 \text{ THz}$ corresponds to $\hbar\omega/k_B \approx 60 \text{ K}$, providing thermodynamic consistency between phonon energy and activation temperature.

NOT REPLACEMENT

SM treats LENR activation thresholds as empirical fit parameters per material system. UQFF supplies a single universal value rooted in the icosahedral integer primitive A_5 .

Reference

- Source: PAPER_1072 SCm Activation Function
- Related: PAPER_1373 (A_5 in Hayflick/IMF), PAPER_1141 (Rossi LENR variants), $\omega_{SCm} = 1.25$ THz
- Calculator dispatch: `calculate_paradox({"paradox": "t_scm_activation_threshold"})`

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